

TrackDancer — CAN Wiring Sheet v3 (transceiver-swap)

SUPERSEDES all earlier wiring sheets (the 3.3V-both and 5V-zener versions). Reprint — wiring changed.

Everything runs at 3.3V — no level shifting anywhere. Check each box as you land it. Both systems POWERED OFF.

A. Teensy side — SN65HVD230 #1 (native FlexCAN, NO SPI)

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|-----------------------------|-----------------|-----------------|------------|
| ☐ 1. | VCC (module) | → 3.3V (Teensy) | RED wire |
| ☐ 2. | GND | → GND (Teensy) | BLACK wire |
| ☐ 3. | CAN TX (module) | → Teensy pin 22 | = CAN1 TX |
| ☐ 4. | CAN RX (module) | → Teensy pin 23 | = CAN1 RX |

B. Pi side — MCP2515 module @ 3.3V (SPI native — every line direct, no shifting)

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|------------------------------|--------------|-----------------|------------|
| ☐ 5. | VCC (module) | → 3V3 (cobbler) | RED wire |
| ☐ 6. | GND | → GND (cobbler) | BLACK wire |
| ☐ 7. | SI | → MOSI | |
| ☐ 8. | SO | → MISO | |
| ☐ 9. | SCK | → SCLK | |
| ☐ 10. | CS | → CE0 | |
| ☐ 11. | INT | → #25 | |

MCP2515 is happy at 3.3V; SPI matches the Pi natively. This is why no dividers/zeners are needed.

C. Transceiver transplant — SN65HVD230 #2 onto the Pi's MCP2515 module

FIRST: desolder & remove the TJA1050 (the small 8-pin chip) from the blue module. Fine-pitch SMD work.

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|------------------------------|-------------------------------|-----------------|------------|
| ☐ 12. | SN65HVD230 #2 3.3V | → 3V3 (cobbler) | RED wire |
| ☐ 13. | SN65HVD230 #2 GND | → GND (cobbler) | BLACK wire |
| ☐ 14. | MCP2515 TXCAN (TJA1050 pad 1) | → #2 CAN TX | |
| ☐ 15. | MCP2515 RXCAN (TJA1050 pad 4) | → #2 CAN RX | |

TJA1050 is SOIC-8: pin 1 = TXD (MCP2515 TXCAN), pin 4 = RXD (MCP2515 RXCAN). Tap those pads after removal.

The blue module's OWN screw-terminal CANH/CANL are now dead (they fed the removed chip) — do not use them.

D. The bus — transceiver to transceiver

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|------------------------------|--------------------|-----------------|--------------------|
| ☐ 16. | SN65HVD230 #1 CANH | → #2 CANH | |
| ☐ 17. | SN65HVD230 #1 CANL | → #2 CANL | |
| ☐ 18. | GND (Teensy brd) | → GND (cobbler) | BLACK — shared ref |

FINAL CHECKS — do these before power-on

- ☐ TJA1050 fully removed — no solder bridging its old pads to anything else
- ☐ MCP2515 module VCC on 3V3 (NOT 5V — this build is all-3.3V)
- ☐ TXCAN→CAN TX and RXCAN→CAN RX not swapped (same rule as the Teensy end)
- ☐ Teensy end: CAN TX→22, CAN RX→23 (not swapped)
- ☐ Both SN65HVD230 boards carry their onboard 120Ω (the '121' resistor) = both ends terminated
- ☐ Multimeter, power OFF: CANH-CANL reads ~60Ω
(two 120Ω in parallel; 120Ω alone means only one end is terminated)

Say each connection out loud as you land it. Slow is fast. — TrackDancer bench sheet v3